

COMPARATIVE BEHAVIORAL STUDY · 5 MODELS · 2 TOPICS · 3 PROMPT TYPES

Different models. *Same ideas.*

We tested 5 LLMs on 2 topics with 3 different prompt types. No matter how we asked, the ideas kept falling into the same handful of patterns.

GPT

Gemini

DeepSeek

Claude

Grok

"Change the model, change the prompt, change the topic. The wording shifts, but the underlying idea structure stays basically the same."

30

conditions tested across models, prompts and topics

6

idea patterns that kept reappearing across every condition

100%

of models produced at least one of the 6 core patterns

Model × Prompt × Pattern

Each cell shows what pattern dominated and what the model actually said. Switch between the two topics to see how consistent this is.

	ELDERLY MOBILITY	EDUCATION	
	OPEN PROMPT	CONSTRAINED	CROSS-DOMAIN
GPT	Optimization + Tech + Platform "Adaptive smart crosswalks", "navigation app", "mobility pods" TECH OVERLAY	Physical Design + Social Systems "slow-lane streets", "walking hosts", "bench libraries" SYSTEM REDESIGN	Network Systems via Analogy "mycelium support hubs", "capillary paths" BIOLOGICAL → NETWORK
Gemini	Tech + Monitoring + Optimization "adaptive pavement sensors", "AI walkers", "RFID signals" HIGH COMPLEXITY TECH	Mechanical + Environmental Design "pneumatic benches", "visual anchoring paths", "handrail networks" ANALOG SYSTEMS	Biological → Engineered Systems "peristaltic bridges", "bioluminescent paths" BIO → INFRASTRUCTURE
DeepSeek	Infrastructure + Service Systems "micro-hubs", "mobility concierge", "rolling lockers" SERVICE LAYER	Low-tech + Social Coordination "walking tax", "bell-crank system", "shared backpacks" POLICY + COMMUNITY	Biological Optimization Systems "slime mold routing", "leaf vein networks" EFFICIENCY ANALOGY
Claude	Balanced Tech + Social + Infrastructure "smart signals", "micro-shuttles", "mobility buddy" HUMAN-CENTERED	Social + Spatial Systems "walking guilds", "barter schemes", "slow lanes" HUMAN COORDINATION	Natural Systems → Urban Flows "tidal streets", "mycelium nodes" FLOW-BASED DESIGN
Grok	Disruption + Lateral Tech "skateable ramps for seniors", "gamified step counters", "crowd-routed paths" CONTRARIAN TECH	Community Infrastructure + Humor "pub-stop benches", "neighbourhood chore swaps", "gossip-walk networks" SOCIAL + PLAYFUL	Biological Swarm → Crowd Systems "ant-trail sidewalks", "murmuration crossing signals" EMERGENT FLOWS

Different inputs, same output types.

Across all 5 models, all 3 prompt types, and both topics, the ideas kept landing in the same 6 structural categories. The examples sound different but they are doing the same conceptual work.

INPUT DIVERSITY

GPT · OPEN · MOBILITY

"adaptive smart crosswalks", "mobility pods"

GEMINI · CONSTRAINED · MOBILITY

"pneumatic benches", "handrail networks"

DEEPSEEK · CROSS-DOMAIN · EDUCATION

"immune memory learning", "ecosystem phases"

CLAUDE · OPEN · EDUCATION

"learning companions", "project-based learning"

GPT · CROSS-DOMAIN · EDUCATION

"mycelium learning", "swarm classrooms"

GEMINI · OPEN · MOBILITY

"AI walkers", "RFID signals"

GROK · CROSS-DOMAIN · EDUCATION

"speedrun learning", "evolutionary curriculum"

GROK · CONSTRAINED · MOBILITY

"gossip-walk networks", "pub-stop benches"

STRUCTURAL OUTPUT

Optimization

Core baseline · all models · all prompts

Tech Integration

Dominant in open · removed by constraint

Platform / System

Appears in ALL conditions

Social Systems

Replaces tech when constrained

Environmental Design

Consistent fallback pattern

Network / Flow Systems

Dominant in cross-domain translation

Which patterns showed up, and how often

Some patterns appeared in every single condition. Others only came out when prompts were open-ended, and disappeared when we added constraints.



What actually changes, and what does not

The model and the prompt do have an effect, just not on the core idea structure. Here is what shifts and what stays fixed.

● WHAT CHANGES

Model identity

Changes detail, tone, complexity, confidence level

Prompt type

Changes language, framing, and domain vocabulary

Surface metaphors

Biological, mechanical or digital, but always the same underlying logic

Tech presence

Drops out when you remove it from the prompt, comes back when you don't

● WHAT STAYS CONSTANT

Core idea structure

Always system-based, regardless of framing

Pattern types

The same 6 show up every time, across every model

Optimization drive

Every model defaults to it, no matter the topic

Network logic

Comes out reliably whenever analogies are used

WHAT THIS MEANS

"The models sound different. But when you look at the structure of what they are generating, they are all working from the same small set of moves."

This started as a prompt experiment. It turned into something more useful: a map of how LLMs actually think about problems. The variety is mostly in the language, not the logic.